Guruprasad
Analog Design Engineer
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Preofessional Summary

Analog Design Engineer with hands-on experience in LDO and power-management circuits in TI CMOS processes. Worked on Texas Instruments client projects involving block-level LDO design support and simulation, PVT and Monte- Carlo analysis, trimming, aging, and post-layout simulations. Experienced in silicon/IP-level characterization of non- volatile memories, VCO, HVGEN, and EEPROM IPs, including testbench development without data sheets. Strong back- ground in analog fundamentals, PMIC blocks, and Cadence Spectre-based design flow.

Core skills

Analog Blocks: LDO, Bandgap, Op-Amps, Comparators, POR, Voltage References

PMIC Concepts: Stability, PSRR, Load Transient, Compensation, Trimming

Verification: PVT, Monte-Carlo, Aging, Burn-in, Corner Analysis

Simulation & Analysis: AC, Transient, Noise, Stability analysis

EDA Tools: Cadence Virtuoso, Spectre, ADE

Technology: TI pdks

Texas Instruments : Client Projects (via KarMic Design)

1. **NMOS LDO Design:**
   - Bias circuit modification and block-level design support and verification of NMOS LDO.
   - PVT, Monte-Carlo, aging, and burn-in simulations
   - Line/load regulation, PSRR, stability, startup, and load transient analysis
   - Resistor trimming and post-layout simulation support

2. **Embedded Memory & NVM IP Characterization**
   - Worked on EEPROM, MTP, and NVM IPs across multiple TI technologies.
   - Developed testbenches for IDDQ, program/read currents, margin testing.
   - External IREF margin characterization across PVT and Monte-Carlo (up to 1000 samples per corner)
   - Post-burn and post-bake current analysis and data extraction.

3. **High-Voltage Generator & VCO IP Testing**
   - HVGENMINILV / HVGENMINIMVAL IP characterization.
   - Designed testbenches to measure IDDQ, frequency vs control voltage.
   - Verified VCO performance across temperature and process corners.
   - Supported performance improvement through circuit-level modifications.